# What's Good about Distributed Databases?

Distributed databases give deeper insights into data, and ensure fault tolerance and data integrity. We take a look at the computational components of distributed databases, as well as the relationship between these databases and polyglot persistence.

As per Wikipedia: "A distributed database is a database in which data is stored across different physical locations. It may be stored in multiple computers located in the same physical location (e.g., a data centre); or may be dispersed over a network of interconnected computers. Unlike parallel systems, in which the processors are tightly coupled and constitute a single database system, a distributed database system consists of loosely coupled sites that share no physical components."

The need for insight from data has made it necessary to store and process increasingly large and diverse data sets, and ensure high availability, scalability, and fault tolerance.

Let's begin by exploring the key stages in the evolution of distributed databases.

***Centralised databases:*** In the early days of computing, databases were typically centralised, with all data stored and managed on a single machine. This approach limited scalability and created single points of failure.

***Client-server model:*** With the advent of networking and the client-server model, databases became more distributed. Multiple client machines could connect to a centralised database server to access and manipulate data. While this improved scalability to some extent, it still relied on a single server for data storage and processing.

***Replication:*** To enhance fault tolerance and availability, replication techniques were introduced. Data was replicated across multiple database servers, allowing for redundancy and increased reliability. However, this approach often faced challenges in maintaining consistency between replicas.

***Partitioning and sharding:*** As data sets continued to grow, databases started employing partitioning and sharding techniques. Partitioning involves dividing the data into smaller subsets and storing them across multiple servers. Sharding, a type of partitioning, distributes data based on specific criteria (e.g., by customer ID or geographical location). These techniques improve scalability and performance by enabling parallel processing of queries across multiple servers.

***Peer-to-peer databases:*** Peer-to-peer (P2P) databases emerged as an alternative to the client-server model. In P2P databases, all participating nodes can act as both clients and servers. Each node contributes storage and processing capabilities, forming a decentralised network. P2P databases offer greater fault tolerance and scalability, as there is no single point of failure.

***NoSQL and new database models***: NoSQL databases, such as document-oriented, key-value, and columnar databases, introduced new data models that better suited distributed environments. These databases prioritise scalability and performance by relaxing